Protopasta

Filament by **Protoplant**

CDP1xxxx
Technical Data Sheet
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Protopasta **Conductive PLA** is a Natureworks INGEO based PLA and carbon black composite designed for printing projects requiring electrical conductivity. Wearable electronics, capacitive touch sensors, and lighting LEDs are some examples of prints using Conductive PLA. With easy printing and reliable conductivity, this material is perfectly suited for these applications.

For quality control during production, we measure the resistance of 10 cm of 1.75 mm filament. We use a custom tool to apply a constant force where the filament contacts the 3 mm polished pin terminals. Below are our QC ranges for this test method for our conductive and dissipative materials. The Ωcm results for the conductive PLA are from using the equation ρ=RA/L, where ρ is the volume resistivity (Ωcm), R is measured resistance, A is cross sectional area of the filament, and L is length.

Material Under Test	**Result Range**	**Meter Used**
Protopasta Conductive PLA	2.0 kΩ - 3.5 kΩ (4.8 Ωcm - 8.4 Ωcm)	Standard DMM
Protopasta Dissipative Materials	100 kΩ - 100,000 kΩ or (10^5 Ω - 10^8 Ω)	DESCO 19786

Formulated and compounded in-house, Protopasta Conductive PLA has been designed for maximum conductivity while maintaining good mechanical properties and trouble free printing.

To verify printed part volume resistivity and to determine variation from orientation and temperature we printed 4 mm x 4 mm x 120 mm bars. A Bambu P1S with AMS and .4 mm nozzle was used to print these parts at a maximum of 8 mm^3/s with 100% infill. The resistance was measured using a DMM and the test fixture described above. The calculated value for volume resistivity is again found using ρ=RA/L.

Volume Resistivity of Printed Part (For Reference Only, Batch and Print Parameter Dependent)

Print Orientation	**Print Temperature (˚C)**	**Measured Resistance (Ω)**	**Calculated Volume Resistivity (Ωcm)**
Flat (X-Y)	210	1200	19.2
Flat (X-Y)	230	900	14.4
Vertical (Z)	210	2100	33.6
Vertical (Z)	230	1700	27.2

Please note that this is an example only and results will vary depending on where in our QC range the batch tests, what your print and geometry parameters are, and your environmental conditions.

Recommended Printing Parameters

Print Temperature	210 - 230 °C
Build Plate temperature	60 °C
Print Speed (Volume Flow)	Up To 8 mm^3/s
Build Plate Preparation	PEI or similar w/ Magigoo or glue stick for release
Nozzle	.4 mm or larger
Abrasive	Non Abrasive
Drying	Generally not required. 55 °C for 6 hrs
AMS Compatibility	Compatible

Processing Notes

Carbon black based conductive products are extremely dark and generally require more purge to change material than other colors.

Electrically conductive polymer composites are sensitive to print orientation, temperature, infill, part geometry, and print speed. Users must test suitability for their specific application.

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